姓名: SOLUTION　　葉均承　　應數一線性代數

學號: _____　　**Quiz 8**　　考試日期: 2026/11/4

不可使用手機、計算器，禁止作弊!

1. (10 points) Check if $T : \mathbb{R}^2 \to \mathbb{R}^2$ is an linear transformation, where $T([x, y]) = [x^2, y]$.

Solution：

Let $\vec{u} = [2, 1]$ and $c = 3 \circ T(3[2, 1]) = T([6, 3]) = [6^2, 3] = [36, 3] \circ$ However, $3T([2, 1]) = 3[2^2, 1] = 3[4, 1] = [12, 3] \circ$ Since $T(3\vec{u}) \neq 3T(\vec{u})$, thus T is NOT an linear transformation.

2. (15 points) Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be a linear transformation such that:

$$T([1, 0, 0]) = [2, -1], \quad T([0, 1, 0]) = [3, 4], \quad T([0, 0, 1]) = [0, 5]$$

(a) Find the standard matrix A for this linear transformation T.

Answer: $A = \begin{bmatrix} 2 & 3 & 0 \\ -1 & 4 & 5 \end{bmatrix}$

(b) Evaluate $T([2, -1, 3])$.

Answer: $T([2, -1, 3]) =$ $[1, 9]$

Solution：

Since A is the standard matrix, we can compute $T(\vec{x}) = A\vec{x}$.

$$T([2, -1, 3]) = \begin{bmatrix} 2 & 3 & 0 \\ -1 & 4 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2(2) + 3(-1) + 0(3) \\ -1(2) + 4(-1) + 5(3) \end{bmatrix} = \begin{bmatrix} 1 \\ 9 \end{bmatrix}$$

3. (15 points) Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation that rotates vectors counterclockwise by an angle of $\frac{\pi}{3}$ $(60°)$.

   (a) Find the standard matrix representation A of the linear transformation T.

   Answer: $A = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$

   Solution：

   The standard matrix for a counterclockwise rotation by θ is $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$.

   Plugging in $\theta = \frac{\pi}{3}$, we know $\cos(60°) = \frac{1}{2}$ and $\sin(60°) = \frac{\sqrt{3}}{2}$.

   Thus, $A = \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}$.

   (b) Find the image of the vector $\vec{v} = [2, 2\sqrt{3}]$ under the transformation T.

   Answer: $T(\vec{v}) = [-2, 2\sqrt{3}]$

   Solution：

   Multiply the standard matrix A by the vector $\vec{v}$:

$$T([2, 2\sqrt{3}]) = \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix} \begin{bmatrix} 2 \\ 2\sqrt{3} \end{bmatrix} = \begin{bmatrix} \frac{1}{2}(2) + \left(-\frac{\sqrt{3}}{2}\right)(2\sqrt{3}) \\ \frac{\sqrt{3}}{2}(2) + \frac{1}{2}(2\sqrt{3}) \end{bmatrix} = \begin{bmatrix} 1 - 3 \\ \sqrt{3} + \sqrt{3} \end{bmatrix} = \begin{bmatrix} -2 \\ 2\sqrt{3} \end{bmatrix}$$